

PES UNIVERSITY



Department of Computer Science & Engineering

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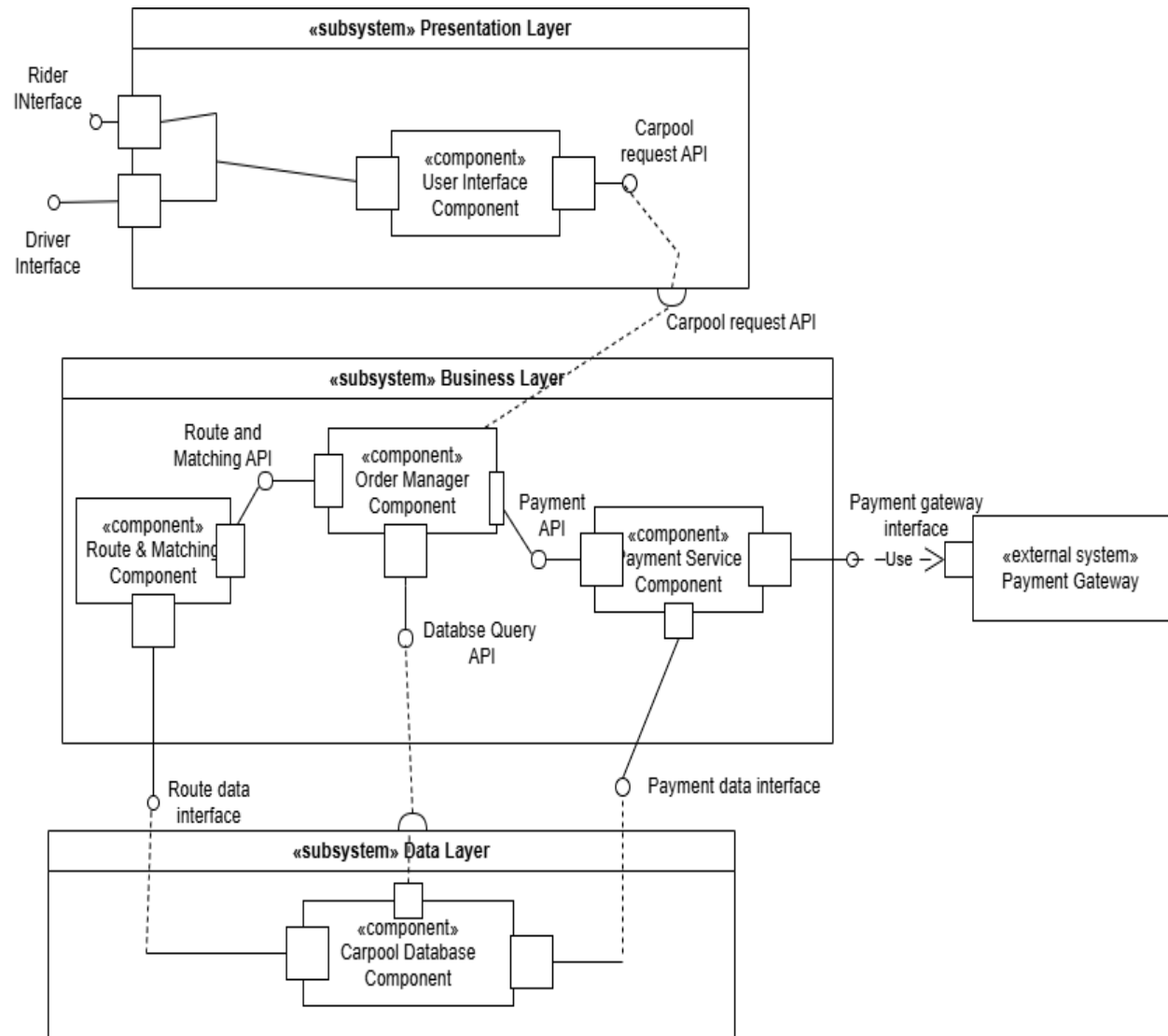
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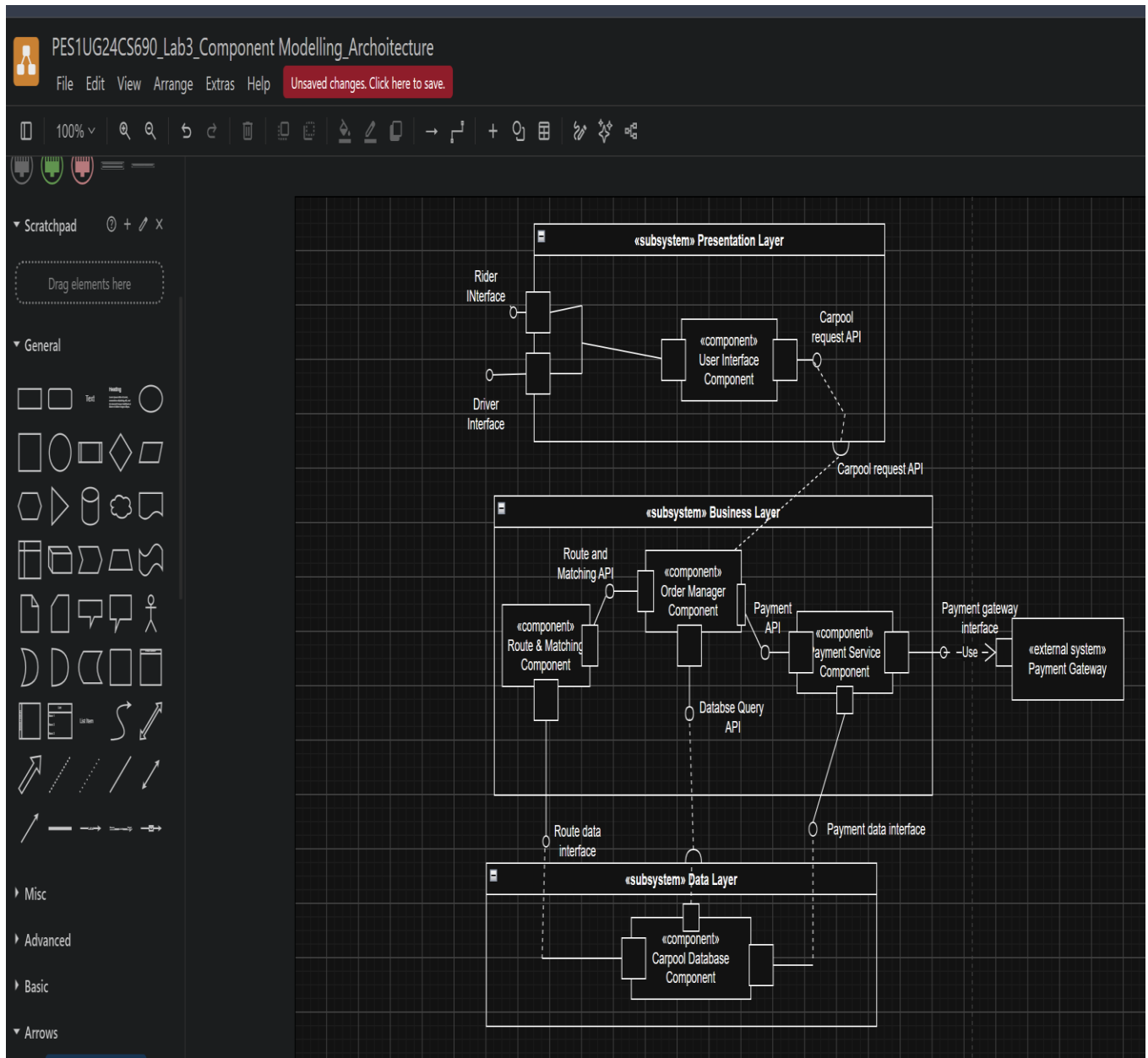
**Lab 3: Component Modelling and Architecture
(Carpooling)**

Name: Shaizan Ulla Shariff

SRN: PES1UG24CS690

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Architecture Selection

I have chosen layered architecture for my Carpooling system.

Architectural Choice

The system is divided into three main layers as presentation, business, and data. The presentation layer deals with the interaction between the riders, drivers, and the system. The business layer handles the main operations such as finding suitable riders, route and match management, and payment and expense processing. The data layer is responsible for storing information such as users, routes, trips, matches, and payment details.

Reason 1 – Separation of Responsibilities

The system has different tasks that need to be handled separately. For eg, users enter their pickup and destination, the system works out suitable routes and matches riders and drivers, and payment and expense details are processed separately. Dividing these tasks into layers makes the system easier to understand and keeps each part focused on its own responsibility.

Reason 2 – Easier Modification

Different parts of the system may need to be changed later. For eg, the user interface, matching logic, or payment processing may be updated independently. Since these functions are separated into different layers and components, making a change in one area should have less effect on the other parts of the system.

Security Advantage

The business layer provides a controlled point between the user interface and sensitive services such as the database and payment service. User requests can be checked before they reach these services. This helps prevent the presentation layer from directly accessing sensitive data or payment operations.

Performance Benefit

Separating the business logic from data storage makes it easier to handle frequently used operations such as route calculation and rider matching through dedicated components. At the same time, database-related operations remain within the Data Layer. This separation makes it easier to optimize these operations individually when required.